

NBD1100 Built Environment Communication and Skills 3D Printing Purpose and Type

3D Printing Presentation 1 - by Shiran Geng

What is 3D Printing?

3D printing is a manufacturing process that allows you to create a physical object from a digital model. This enables you to produce complex yet functional shapes cheaply and without needing a detailed understanding of manufacturing. This digital-to-physical workflow is known as digital fabrication.

There are two main methods used for digital fabrication within the industry: **Subtractive** and **Additive** processes. **Subtractive** processes remove material from a solid block. An example is a CNC machine which uses a computer-guided drill to sculpt a model from a solid material. 3D Printing is an **Additive** process because it builds a model from the material. A very small computer-controlled nozzle is used to lay down plastic layer by layer.

Today, more industries and design schools are embracing digital fabrication and 3D printing in the design process, using it for anything from design exploration, prototyping, and exhibition models, to both small and large-scale fabrication. Compared to traditional manufacturing techniques, 3D printing and digital fabrication present new and exciting potentials!



Advantages of 3D Printing

Complex Forms	3D Printing enables the fabrication of complex, intricate and detailed designs. Geometries benefit from the direct digital-to-physical workflow.
Iteration & Customisation	Take advantage of the digital fabrication workflow. Digital models can be rapidly manipulated and generated, used as quickly as they can be printed. This can also be used for mass customisation and adaptive designs.
Functioning Models	3D printed models can be used as printed; they don't need further processing. This makes them useful as prototypes, functional products or as part of a fabrication process.
Replicable Models	As a computer-controlled process, the same model can be accurately produced to designed tolerances - models can be hot-swapped to present an array of options.
Accessibility	3D printing outcomes are mostly determined before they even reach the printer; digital model-making instead of physical model-making. It requires less oversight compared model-making via hand or laser cutting as the machine is automated, making it an accessible process.

What is the modelling purpose? Is 3D Printing the one for you?

Implement 3D Printing	Consider other Fabrication Methods
Express form. Models with any non-conventional design form can be difficult to represent using traditional fabrication methods, especially complex and intricate geometry, such as double curvatures.	**Planar, rectilinear, flat geometry**. 3D printing is not a 'flat' process nor material, such as wood or MDF. It has the capacity to print non-planar surfaces with much more ease and efficiency than other fabrication methods, when keeping the design guidelines and constraints in mind. See if you can economise on these properties and push the limitations of 3D printing.
Surface detail to communicate some level of texture. Communication of materiality will need additional process.	**Express materiality & colour.** 3D prints show the form accurately, but they don't accurately portray the design's intended material. Most 3D prints will only use one material while printing. Consider additional finishing and post-processing.
Small to medium scale. 3D printers don't have infinite range, so even though you can slice up your print, the pieces at least are going to have to stay inside the boundaries.	**Large surface area prints.** For example, Massing Models and Site Models, unless complex in shape, take up a lot of surface area on the bed and tend to be flat. This leads to warping and a large amount of material usage, where it would be best to use another technique.
Rapidly iterate and test ideas. The advantages of the digital fabrication workflow, enable you to quickly test design ideas and display a lineage and variety of options.	**Design through model-making.** Due to the automated nature of the digital fabrication process, you will have more design agency throughout the digital fabrication stage (modelling) instead of the physical manufacturing process. To manipulate or make changes to outcomes, you will need to iterate through the digital making process from CAD to print.

Example Model Types – Conceptual Model

A conceptual model allows the designer to develop initial concepts and ideas to develop a form. Concept models can be used to express any number of your project's elements and/or features. The direction you choose to go down is really dependent on where your concept is being derived from.

There should be a certain level of interpretation, and they may never represent the final design, (that's a whole different type of model), but may show a section or chunk that communicates the concept.



A simple model can convey a concept very strongly, eg. a faceted tower that twists.



Conceptual Model exploring discrete assembly formal languages.

Example Model Types – Conceptual Model Parameters

[Scale / Scope] : As an initial abstract model, you will less likely need to represent the 'whole'. Consider the best portion and scale of the whole that your initial concept will be best communicated at. This affects the time of the print and how effectively your concept is presented.

[Supports] : Minimise support as much as possible and therefore printing time. Concept creation is solely about experimenting and testing. If you have chosen 3D printing as your method for the exploration of an idea, ensure you are optimising your geometry as much as possible to avoid delays in your initial creation process.

[Detail / Resolution] : If your focus is only diagrammatic, keep it simple. The concept creation process may or may not require a high level of resolution or detail, unless that is a critical element driving the concept.

[Quality / Finish] : Unless your concept is intending to represent and express materiality, the finish can be of a lesser quality than presentation models. A looser approach is usually taken to the construction process of the models and are made quickly. However, conceptual models can also be refined for a final and more presentable concept as a final presentation tool.

Example Model Types – Study Model

A study model aims to explore a collection of aspects of a design. Once an idea gains traction, it's time to develop and take it as far forward as it will go. 3D Printing excels in this study workflow as you can quickly iterate through various ideas with the help of digital tools.

The 3D printing digital fabrication process enables you to rapidly iterate, test and adapt your designs. It can be beneficial to keep each iteration and development, as this will later help communicate your refined idea.

Study models allow you to be selective about the focus of the model: you can look at the entire design or dial in on design elements. You might be looking at a handful of overall qualities from the geometry.

These may be at the architectural scale (silhouette, composition, light/shadow, space) or at the product scale (ergonomics, scale). You may also study very detailed aspects of the design and how particular parts or textures come together.



Study Model testing the design limitations of 45 degree draft angles.

Example Model Types – Conceptual Model Parameters

[Scale / Scope] : By identifying the purpose of the study model first; you can be selective about how much of the model is necessary to print. Consider the portion and scale of the whole that your study exploration will be best communicated at.

[Detail / Resolution] : The digital model should only be as detailed as necessary to express your intended qualities.

[Quality / Finish] : The digital model resolution and model finish should only be as refined as necessary to express your intended qualities and study exploration.

Example Model Types – Presentation Model

A presentation model aims to communicate a resolved and synthesized design idea to communicate the full conceptualization of the project. The final model does not need to literally be represented as a whole. It can be represented as a part of the whole, as similarly described for both the Concept and Study Models.

Mixed Fabrication methods

Producing composite final Presentation Models by mixing 3D printing with other forms of fabrication, such as laser-cutting, is a common technique. This will save time and allow you to present your project as a whole, incorporating varying levels of detail, materiality and specificity depending on the focus of each element in the model.

For example, laser-cut materials have minimal texture and that may tie into your choice of materiality for the context. Keep the limits of 3D Printing in mind, it excels at accurately communicating form and sometimes texture depending on the scale.



Final Presentation Model situated within the site.

Example Model Types – Presentation Model Parameters

[Scale / Scope] : By identifying the purpose of the presentation model first; you can be selective about how much of the model is necessary to print. Presentations model can show both the whole or a detailed selection of your idea.

[Detail / Resolution] : The digital model should only be as detailed as necessary to express your intended qualities. You may need to split up your model if you require presentation-quality detailing. A completely 3D printed model will usually need to be split up into parts. This will maximise the print quality for visible surfaces and minimise the time of the print as well.

[Finishing / Post Processing] : Presentation models intend to, as accurately as possible, represent the final proposed outcome. If your concept is intending to represent and express materiality, the finish may need to be of a higher and more expressive quality. Be sure to check out our Post Processing and finishing techniques.

Example Model Types – Function Model

3D Printing can also be used to directly fabricate functioning models, that serves a purpose and is intended to be used coming off the printer.

A functional Model does not require any abstractions (compared to Representational Models) as the design should be modelled directly for printing. The following are parameters that are generally universal for functional models.

While you should consider all parameters when modelling for 3D printing; these may require more attention.



3D Printed parts of models.

Example Model Types – Function Model Parameters

[Detail / Resolution] Detail will be limited to the printer's resolution.

[Orientation] & **[Supports]** Will influence the overall finish as well as the strength of the model.

[Post processing] May be required to finish the model. **[Watertight]** True water-tightness might need to be considered, as 3D prints may be porous.

[Large Models] If a model is too large for a printer's volume, then you may need to consider splitting it up.

If the model requires any additional hardware then this will need to be considered in the design of receivers and notches; closely related to **[tolerance]** as well.



Example Model Types – Static Model

Static models encompass most models that come off the 3D printer ready to use. As long as you consider the overall guidelines and the key general parameters, these prints are straight-forward design-to-print.

These prints can range from prototypes, production models, engineering parts and archival models.

Example Model Types – Dynamic Model

Dynamic models feature moving parts: engineering models and gears, kinetic systems etc. Moving parts can be printed individually and assembled later. However, an advantage of 3D printing is that embedded systems can be printed as a single unit without further assembly.

To ensure that a dynamic 3D print model functions as intended, the digital model should be precise and will need to account for the tolerances of the 3D printer.

Parameters:

[Tolerances] are very important to ensure that the parts of the model can work together.

[Detail/Resolution] will also limit how small these moving parts may be.

[Orientation] & **[Strength]** should also be considered to ensure parts are strong enough to stand up to any forces in the model.

Single embedded models can be held together by **[minimal support material]**, or to take advantage of the printer properties and print settings; **[Supports]** to print a dynamic model without assembly required.



3D printing to create a model that is fully functioning and simulates a human lung.

Reference